

PRACTICAL : 3**Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind****Problem Definition:**

Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first Responsiveness.

A) Theory**1. How does the layout adapt for mobile, tablet, and desktop screens?****Answer:**

The layout changes according to the screen size using responsive design. On mobile screens, the content is arranged in a single column. On tablets, the layout adjusts to provide more space. On desktop screens, the content is displayed in multiple columns for a better user experience.

2. Which layout technique is used and why?**Answer:**

Flexbox and CSS Grid are used for the layout. Flexbox helps align items in rows or columns, while CSS Grid is useful for arranging content in a structured layout. These techniques make the website responsive and easy to manage.

3. Are typography, spacing, colors, and contrast readable?**Answer:**

Yes. Simple fonts, proper spacing, readable font sizes, and good color contrast are used. This makes the content easy to read and improves the overall user experience.

4. Is the framework used consistently without unnecessary inline CSS?

Answer:

Yes. The same CSS framework and external stylesheet are used across all pages. Inline CSS is avoided as much as possible to keep the code clean, reusable, and easy to maintain.

B) code :

1) style.css

```
/* Body */  
  
body{  
  
    background-color: lightgray;  
  
    font-family: Arial, sans-serif;  
  
    text-align: center;  
  
}  
  
h1{  
  
    color: blue;  
  
}  
  
title{  
  
    color: blue;  
  
    text-align: center;  
  
    font-weight: bold;  
  
}  
  
/* Table */
```

```
table{

    width: 80%;

    margin: auto;

    border-collapse: collapse;

    background-color: white;

}

th{

    background-color: skyblue;

    color: black;

    padding: 10px;

    border: 1px solid black;

}

/* Table Data */

td{

    padding: 10px;

    border: 1px solid black;

}

/* Text Box */

input[type="text"]{

    padding: 8px;

    width: 200px;
```

```
}

/* Buttons */

input[type="submit"],
input[type="reset"]{

    padding: 8px 15px;

    background-color: blue;

    color: white;

    border: none;

}

/* Links */

a{

    text-decoration: none;

    color: blue;

    margin: 10px;

}

a:hover{

    color: red;

}
```

2)about.css

```
body{

    background-color: #eeeeee;

    font-family: Arial;

}


table{

    width: 100%;

    background-color: white;

}


td{

    padding: 10px;

}


img{

    border: 2px solid black;

}


a{

    color: blue;

    text-decoration: none;

    padding: 5px;

}
```

```
a:hover{  
  
    color: red;  
  
}  
  
h1{  
  
    color: navy;  
  
    text-align: center;  
  
}  
  
h2{  
  
    color: green;  
  
}  
  
ul{  
  
    background-color: white;  
  
    padding: 10px;  
  
}  
  
p{  
  
    font-size: 16px;  
  
}
```

```
hr{  
  
    border: 1px solid gray;  
  
}
```

3)assignment.css

```
body{  
  
    background-color: #f2f2f2;  
  
    font-family: Arial;  
  
}  
  
h1{  
  
    color: darkblue;  
  
}  
  
table{  
  
    width: 80%;  
  
    background-color: white;  
  
    border-collapse: collapse;  
  
}  
  
th{  
  
    background-color: royalblue;  
  
    color: white;
```

```
}

th,td{

    border: 1px solid gray;

    padding: 10px;

    text-align: center;

}

tr:hover{

    background-color: #e6f2ff;

}

label{

    font-weight: bold;

}

input[type="submit"],

input[type="reset"]{

    background-color: royalblue;

    color: white;

    border: none;

    padding: 8px 15px;

}
```

```
input[type="submit"]:hover,  
input[type="reset"]:hover{  
    background-color: darkblue;  
}  
  
a{  
    text-decoration: none;  
    color: blue;  
    margin: 8px;  
}  
  
a:hover{  
    color: red;  
}  
  
hr{  
    width: 80%;  
}
```

4)attendance.css

```
body {  
    font-family: Arial, sans-serif;
```

```
background-color: #f2f2f2;

margin: 20px;

text-align: center;
}

h1 {

color: white;

background-color: #0077cc;

padding: 15px;

border-radius: 8px;
}

table {

width: 90%;

margin: auto;

border-collapse: collapse;

background-color: white;
}

th {
```

```
background-color: #0077cc;

color: white;

padding: 10px;
}

td {

padding: 10px;

border: 1px solid black;
}

tr:nth-child(even) {

background-color: #f9f9f9;
}

tr:hover {

background-color: #d6ecff;
}

label {

font-weight: bold;

font-size: 18px;
}
```

```
input[type="text"] {  
  
    width: 250px;  
  
    padding: 8px;  
  
    border: 1px solid gray;  
  
    border-radius: 5px;  
  
}  
  
input[type="submit"],  
input[type="reset"] {  
  
    background-color: #0077cc;  
  
    color: white;  
  
    border: none;  
  
    padding: 10px 20px;  
  
    margin: 5px;  
  
    border-radius: 5px;  
  
    cursor: pointer;  
  
}  
  
input[type="submit"]:hover,  
input[type="reset"]:hover {  
  
    background-color: #005fa3;  
  
}
```

```
a {  
  
    display: inline-block;  
  
    text-decoration: none;  
  
    color: white;  
  
    background-color: #28a745;  
  
    padding: 10px 15px;  
  
    margin: 5px;  
  
    border-radius: 5px;  
  
}  
  
a:hover {  
  
    background-color: #1e7e34;  
  
}
```

5)contact.css

```
body {  
  
    font-family: Arial, sans-serif;  
  
    background-color: #f2f2f2;  
  
}  
  
h2 {
```

```
text-align: center;

color: blue;
}

form {

width: 300px;

margin: auto;

background-color: white;

padding: 20px;

border: 1px solid gray;
}

label {

font-weight: bold;
}

input,
textarea {

width: 100%;

padding: 8px;

margin-top: 5px;

margin-bottom: 10px;
```

```
border: 1px solid gray;

box-sizing: border-box;

}

textarea {

    height: 80px;

}

input[type="submit"],
input[type="reset"] {

    width: 48%;

    background-color: blue;

    color: white;

    border: none;

    padding: 10px;

    cursor: pointer;

}

input[type="reset"] {

    background-color: gray;

}

input[type="submit"]:hover {
```

```
background-color: darkblue;

}

input[type="reset"]:hover {

    background-color: black;

}
```

6)course.css

```
body {

    font-family: Arial, sans-serif;

    background-color: #f2f2f2;

    margin: 20px;

}

table {

    width: 100%;

    border-collapse: collapse;

}

a {

    text-decoration: none;

    color: blue;

    margin: 5px;
```

```
}

a:hover {

    color: red;

}

h1 {

    color: darkblue;

}

th {

    background-color: blue;

    color: white;

}

th,

td {

    border: 1px solid black;

    padding: 10px;

    text-align: center;

}
```

```
tr:nth-child(even) {  
  
    background-color: #eeeeee;  
  
}  
  
p {  
  
    color: gray;  
  
}
```

7) login.css

```
body {  
  
    font-family: Arial, sans-serif;  
  
    background-color: #f2f2f2;  
  
}  
  
h2 {  
  
    color: blue;  
  
}  
  
input {  
  
    width: 250px;  
  
    padding: 8px;
```

```
margin-top: 5px;

border: 1px solid gray;
}

button {

width: 100px;

padding: 10px;

background-color: blue;

color: white;

border: none;

cursor: pointer;
}

button:hover {

background-color: darkblue;
}

input[type="submit"],

input[type="reset"] {

width: 100px;

padding: 10px;

background-color: green;
```

```
color: white;

border: none;

cursor: pointer;
}

input[type="reset"] {

    background-color: red;
}

input[type="submit"]:hover {

    background-color: darkgreen;
}

input[type="reset"]:hover {

    background-color: darkred;
}

input[type="checkbox"] {

    width: auto;
}

a {
```

```
text-decoration: none;

color: blue;
}

a:hover {

color: red;
}
```

8)profile.css

```
body {

font-family: Arial, sans-serif;

background-color: #f2f2f2;

margin: 20px;
}

h1 {

color: blue;
}
```

```
form {  
  
    width: 400px;  
  
    margin: auto;  
  
    background-color: white;  
  
    padding: 20px;  
  
    border: 1px solid gray;  
  
}
```

```
label {  
  
    font-weight: bold;  
  
}
```

```
input,  
  
select,  
  
textarea {  
  
    width: 100%;  
  
    padding: 8px;  
  
    margin-top: 5px;  
  
    margin-bottom: 10px;
```

```
border: 1px solid gray;

box-sizing: border-box;

}
```

```
input[type="radio"] {

    width: auto;

}
```

```
input[type="file"] {

    width: auto;

}
```

```
textarea {

    height: 80px;

}
```

```
input[type="submit"] {

    background-color: blue;

    color: white;

    border: none;
```

```
        cursor: pointer;

    }

    input[type="submit"]:hover {

        background-color: darkblue;

    }


    a {

        text-decoration: none;

        color: blue;

        margin-right: 10px;

    }

    a:hover {

        color: red;

    }


    hr {

        margin-top: 30px;

    }
```

9) result.html

```
body {  
  
    font-family: Arial, sans-serif;  
  
    background-color: #f2f2f2;  
  
    margin: 20px;  
  
}
```

```
h1 {  
  
    text-align: center;  
  
    color: yellow; ;  
  
}
```

```
form {  
  
    width: 350px;  
  
    margin: auto;  
  
    background-color: white;  
  
    padding: 20px;  
  
    border: 1px solid gray;  
  
}
```

```
label {  
  
    font-weight: bold;  
  
}
```

```
input,  
  
select {  
  
    width: 100%;  
  
    padding: 8px;  
  
    margin-top: 5px;  
  
margin-bottom: 10px;  
  
border: 1px solid gray;  
  
    box-sizing: border-box;  
  
}
```

```
input[type="submit"],  
  
input[type="reset"] {  
  
    width: 48%;  
  
    background-color: blue;  
  
    color: white;  
  
    border: none;
```

```
padding: 10px;

cursor: pointer;

}

input[type="reset"] {

background-color: gray;

}

input[type="submit"]:hover {

background-color: darkblue;

}

input[type="reset"]:hover {

background-color: black;

}


table {

width: 100%;

border-collapse: collapse;

margin-top: 20px;

}
```

```
th {  
  
    background-color: blue;  
  
    color: white;  
  
}
```

```
th,  
  
td {  
  
    border: 1px solid black;  
  
padding: 10px;  
  
text-align: center;  
  
}
```

```
tr:nth-child(even) {  
  
background-color: #eeeeee;  
  
}
```

```
a {  
  
    text-decoration: none;  
  
color: blue;  
  
}
```

```
a:hover {  
  
    color: red;  
  
}
```

10) feedback.css

```
body {  
  
    font-family: Arial, sans-serif;  
  
    background-color: #f2f2f2;  
  
}
```

```
h2  
  
{  
  
text-align: center;  
  
color: red;  
  
}
```

```
form {  
  
    width: 350px;  
  
margin: auto;  
  
    background-color: white;
```

```
padding: 20px;

border: 1px solid gray;

}


label {

    font-weight: bold;

}


input,

select,

textarea

{

    width: 100%;

    padding: 8px;

    margin-top: 5px;

    margin-bottom: 10px;

    border: 1px solid gray;

    box-sizing: border-box;

}
```



```

input[type="submit"]:hover {

    background-color: darkblue;

}

input[type="reset"]:hover {

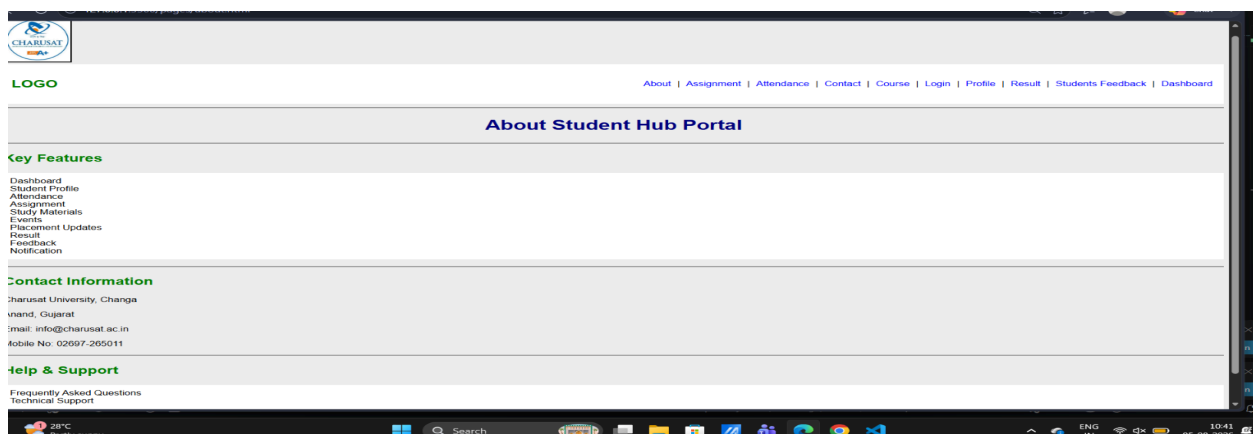
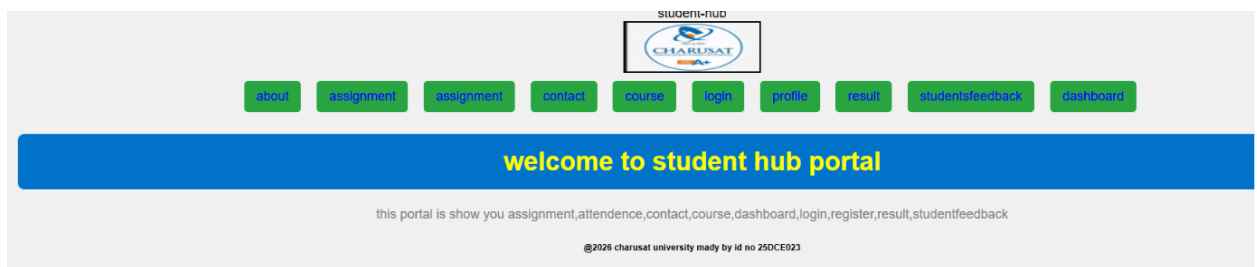
    background-color: black;

}

```

C) Evidence of work

Style.css



student-hub assignment portal

assignment no	assignment name	assignment description	assignment submission date
1	assignment 1	this is assignment 1 description	2024-06-30
2	assignment 2	this is assignment 2 description	2024-07-15
3	assignment 3	this is assignment 3 description	2024-08-01

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student-hub attendance portal

date	subject	total lectures	lectures attended	lecture absent	attendance percentage
2024-06-01	Mathematics	20	18	2	90%
2024-06-02	java	15	12	3	80%
2024-06-03	CN	10	9	1	90%
2024-06-03	DSA	12	8	4	66.67%
2024-06-03	VDF	11	4	7	36.36%

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About Assignment Attendance Contact Course Login Profile Result Students Feedback Dashboard

Available Courses

Course ID	Course Name	Duration	Faculty
ITUC201 FCN	Computer Networking	6 Months	Mr. Nishant Koshl
CEUC-203	Java Programming	6 Months	Mr. Devan Got
ITUC-203	Web Development	6 Months	Ms. Priyanka Padliyar
CSUC-201	Data Structure and Algorithms	6 Months	Mr. Priemal Patel
MSUD-203	Mathematics	6 Months	Mr. Rajesh Savaliya

Student Hub Portal
25dce023@charusat.edu.in

student-hub profile
profile

profile photo
 No file chosen

full name

enrollment number

email

mobile number

date of birth

Gender ☐ male ☐ female ☐ other

course

semester

address

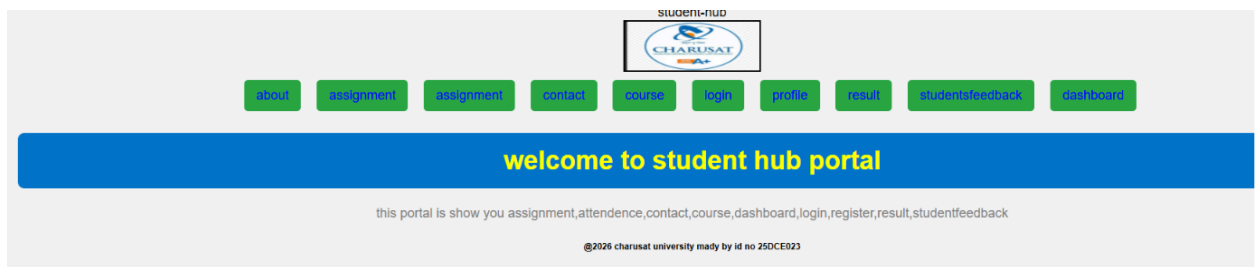
student-hub result portal

enrollment number:

semester:

subject	marks obtained	total marks	result
mathematics	90	100	pass
java	80	100	pass
DSA	85	100	pass
CN	85	100	pass
DSA	85	100	pass

Dashboard



D) Github repository link

[Studenthub/css at main · nency1709/Studenthub](#)

E) problem facing :

While designing the responsive pages, I found it difficult to arrange the layout for different screen sizes. At first, I was confused about when to use CSS Grid and when to use Flexbox. Making the pages look good on mobile, tablet, and desktop required several changes. I also had to adjust spacing and alignment to keep the design consistent. After testing the pages on different screen sizes, I fixed the issues and completed the layout.

F) Learning Outcomes

From this practical, I learned how to create responsive web pages using CSS Grid and Flexbox. I understood the mobile-first approach and how layouts change on different screen sizes. I also learned to maintain proper spacing, alignment, and consistency across all pages. This practical improved my CSS layout and responsive design skills.

G) faculty remarks :